

Chapter 1

POLYNOMIALS

Polynomial:

A sum of finite number of monomials is called a polynomial. Each monomial is called a term of the polynomial.

Monomial:

A monomial is a variable, or a constant, or a product of constant and one or more variables, with the variables having only non-negative integer in exponents.

Example :

$3x^2y$, $-5xy$, and $-7xy^3$ are monomials.
The algebraic expression

$$4y^{-3} \text{ and } \frac{3}{y}$$

are not monomials, because these expressions have not non-negative integer in exponent, and cannot be written as a product of a constant and a variable with a non-negative integer exponent.

Degree of Monomial:

In any monomial the sum of the exponents of the variables is called the degree of monomial.

Example : What are the degrees of the monomials

$$-3x^2y, 7x^3y, -18xy^2$$

Solution:

In algebraic expression $-3x^2y$, the degree of the monomial is 3, because the exponents of x and y are 2 and 1 respectively therefore their sum is $(2 + 1 = 3)$. Similarly the degree of the expressions $7x^3y$ and $-18xy^2$ are 4 and 3 respectively.

Note:

In monomial, the constant is called the numerical coefficient or simply the coefficient of the monomial. $-3x^2y$, $7x^3y$ and $-18xy^2$ are monomials of coefficient -3 , 7 , and -18 respectively.

Multiplication of Monomials:

The process of multiplication is illustrated in the following example:

Example : What is the value of $-5xy^2$, when $x = -2$ and $y = -3$

Solution:

First of all write the coefficient of the monomial, then substitute the value of x and y in monomial. Then evaluate:

$$-5(-2)(-3)^2 = -5(-2)(9) = 90$$

Polynomial:

A sum of a finite number of monomials is called a polynomial. Each monomial in a polynomial is called a term of the polynomial.

Degree of a Polynomial:

The degree of a polynomial is the largest degree of the terms in the polynomial.

What are like terms in a Polynomial?

Terms of polynomial that have exactly the same variables raised to the same powers are called like terms.

Remember: Only like terms in a polynomial can be combined.

Arithmetic Operations on Polynomials:

We use usual law of arithmetic, to add subtract, multiply and divide polynomials.

Addition and Subtraction:

Polynomials are added or subtracted by combining like terms.

Example : $(2x^3 + 3x^2 + 7x + 6) + (4x^2 + 3x - 2) - (5x^2 + 4x)$
 $= 2x^3 + (3x^2 + 4x^2 - 5x^2) + (7x + 3x - 4x) + (6 - 2)$
 $= 2x^3 + 2x^2 + 6x + 4$

The rules for adding like terms are:

Rule 1:

If all the terms are positive in a polynomial, then add their coefficients.

Example : Find the value of $8x^2 + 2x^2 + 7x^2$

Solution: Here we have to increase 8 like things by 2 and 7 like things of the same kind, and aggregate is 17 of each thing.

Rule 2: If all the terms in a polynomial are negative add the coefficient numerically and prefix the minus sign to the sum.

Example : What is the sum of $-4x, -x, -3x$ and $-7x$

Solution: In this example the word sum indicates the aggregate of 4 subtractive quantities of like terms. In this case we have to take away successively 4, 1, 3 and 7 like things, therefore the result is the same as taking away $15(4 + 1 + 3 + 7)$ such things in the aggregate.

$\therefore$ The sum of $-4x, -x, -3x, -7x$ is $-15x$.

Rule 3: If all the terms have not same sign, add together separately the coefficient of all the negative terms and the coefficient of all the positive terms: Then find the difference of those two results, preceded by the sign of the greater, will give the coefficient of the sum required.

Example : Find the sum of $12x^2 - 3x^2 + 15x^2 - 17x^2$

Solution: The sum of the coefficient of positive terms is $12 + 15 = 27$

The sum of the coefficient of negative terms is $3 + 17 = 20$

The difference of these is 7, and the sign of the greater is positive: hence the required sum is $7x^2$.

Multiplication of Monomials: To multiply two simple monomials together, first multiply their coefficients together and prefix their product to the product of the different letters, giving to each letter

an index equal to the sum of the indices that letter has in the separate factors.

Example : What is the product of $5x^2y^3$ and $-3xy^2$

$$\text{Solution: } (5x^2y^3)(-3xy^2) = (5)(-3)(x^2 \times x)(y^3 \times y^2) \\ = -15x^3y^5$$

Note: The product of a monomial by any polynomial is the algebraic sum of the partial products of each term of the polynomial by that monomial.

Multiplication of two Binomials:

The procedure of multiplication of two binomials is illustrated as:

1. Multiply each term of the first binomial by each term of the second.
2. When the terms multiplied together have like signs, prefix to the product the sign +, when unlike prefix -.
3. The algebraical sum of the partial products so formed gives the complete product.

Example : Multiply $(x + 3)$ by $(x - 5)$

$$\text{Solution: } (x + 3)(x - 5) = x(x - 5) + 3(x - 5) \\ = x^2 - 5x + 3x - 15 \\ = x^2 - 2x - 15$$

Example : Find the value of $(x + 2)(x - 3) - (x + 4)(x - 5)$

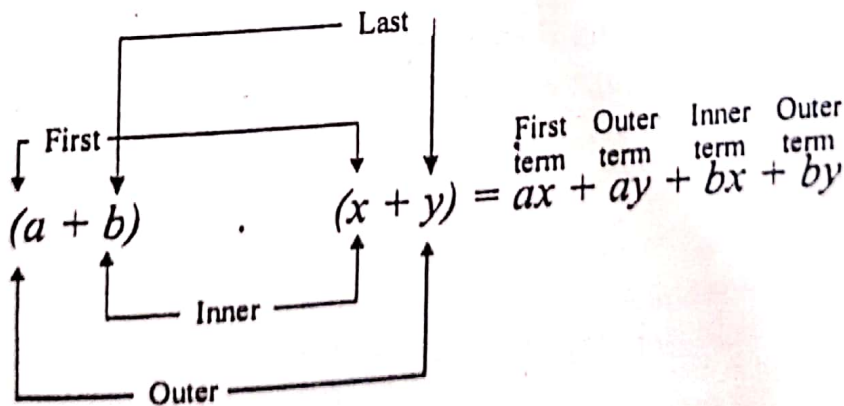
Solution: First of all, multiply both pairs of binomials separately, then subtract the second result from the first.

$$(x + 2)(x - 3) = x(x - 3) + 2(x - 3) = x^2 - 3x + 2x - 6 \\ = x^2 - x - 6$$

$$(x + 4)(x - 5) = x(x - 5) + 4(x - 5) = x^2 - 5x + 4x - 20 \\ = x^2 - x - 20$$

$$\text{Subtracting: } (x^2 - x - 6) - (x^2 - x - 20) = x^2 - x - 6 - x^2 + x + 20 \\ = 14$$

FOIL Method: The product of the two binomials can be computed by the FOIL method. This method is illustrated in the following example



Example : Find the product of $(2x - 3)$ and $(4x + 2)$ using FOIL Method

$$\begin{array}{ccccccc} & & \text{First} & \text{Outer} & \text{Inner} & \text{Last} & \\ (2x - 3)(4x + 2) & = & (2x)(4x) & + & (2x)(2) & + & (-3)(4x) & + & (-3)(2) \\ & = & 8x^2 & + & 4x & - & 12x & - & 6 \\ & = & 8x^2 & - & 8x & - & 6 & \end{array}$$

Important Binomial Formulas:

Following are most important binomial products, those occur frequently in algebra.

1. $(x + y)(x - y) = x^2 - y^2$
2. $(x + y)^2 = x^2 + 2xy + y^2$
3. $(x - y)^2 = x^2 - 2xy + y^2$

Example : Find each of the following products:

- a) $(2a + 3)(2a - 3)$
- b) $(a - 5b)^2$

Solution:

- a) Using formula $(x + y)(x - y) = x^2 - y^2$
Here $(2a + 3)(2a - 3) = (2a)^2 - (3)^2 = 4a^2 - 9$
- b) Using formula $(x - y)^2 = x^2 - 2xy + y^2$
 $(a - 5b)^2 = (a)^2 - 2(a)(5b) + (5b)^2 = a^2 - 10ab + 25b^2$

Example : Given $x + y = 5$, and $x^2 - y^2 = 10$, what is the value of $x - y$?

Solution: Using the fact $(x + y)(x - y) = x^2 - y^2$

$$\begin{aligned} (5)(x - y) &= 10 \\ x - y &= \frac{10}{5} \\ \Rightarrow x - y &= 2 \end{aligned}$$

Example : Find the value of xy , when $(x + y)^2 = 25$ and $x^2 + y^2 = 3$

Solution: We know $(x + y)^2 = x^2 + 2xy + y^2$

We can write $(x + y)^2 = (x^2 + y^2) + 2xy$

Substituting the value of $(x + y)^2$, and $x^2 + y^2$ in above

$$\begin{aligned} 25 &= (3) + 2xy \\ 22 &= 2xy \\ xy &= \frac{22}{2} \\ xy &= 11 \end{aligned}$$

Division of Polynomial by Monomial:

Division is the inverse of multiplication. The object of division is to find out the quantity called quotient.

Thus $\frac{\text{divided}}{\text{divisor}} = \text{quotient}$

To divide a monomial by a monomial, use distributive law, the index of each letter in the quotient is obtained by subtracting the index of that letter in the divisor from that in the divided. To the result so obtained prefix its proper sign the quotient of the divided by that of divisor.

To divide a polynomial by a monomial, divide each term separately by that monomial, and take the algebraic sum of the partial quotient so obtained.

Example 1: What is the quotient when $-4x^2y$ is divided by $2x$.

Solution: The quotient $= \frac{-4x^2y}{2x} = -2xy$

Example 2: Divide $12x^3 - 6x^2 - 9x$ by $3x$

Solution: $\frac{12x^3}{3x} - \frac{6x^2}{3x} - \frac{9x}{3x} = 4x^2 - 2x - 3$

Evaluating a Polynomial:

To evaluate a polynomial, substitute the given value(s) for the variable(s) and then perform the given operation.

Example : If $x = 3$, $y = -7$ and $z = -2$, find the value of $x^2 - 26y + 17z$

Solution: $x^2 - 26y + 17z = (3)^2 - (26)(-7) + 17(-2)$
 $= 9 + 182 - 34 = 157$

FACTORISING POLYNOMIALS:

Writing a polynomial as a product of polynomials of lower degree is called factoring. When each of the terms which compose a polynomial is divisible by a common factor, the polynomial may be simplified by dividing each term separately by this factor, and enclosing the quotient within brackets; the common factor being placed outside as a coefficient.

Example 1: Resolve into factors $4x^2 - 20x$

Solution: The terms of the polynomials $4x^2 - 20x$ have a common factor $4x$;

$$\therefore 4x^2 - 20x = 4x(x - 5)$$

Example 2: Resolve into factors $x^2 - sx + tx - st$

Solution: We see that the first two terms contain a common factor x , and the last two terms a common factor t , we enclose the first two terms in one bracket, and the last in another. Thus,

$$\begin{aligned} x^2 - sx + tx - st &= (x^2 - sx) + (tx - st) \\ &= x(x - s) + t(x - s); \text{ take } (x - s) \text{ common} \\ &= (x - s)(x + t) \end{aligned}$$

Factorising Quadratic Trinomials: Some trinomials of the form $x^2 + bx + c$ can be factorized by trial and error procedure. This method is the reverse of the FOIL method. This is illustrated in the following example.

Example : Consider the following binomial expansion:

$$\begin{aligned}(x+5)(x+6) &= x(x+6) + 5(x+6) \\ &= x^2 + 6x + 5x + 30 \\ &= x^2 + 11x + 30 \\ &= (x+5)(x+6)\end{aligned}$$

Notice at $11 = 5 + 6$ and $30 = 5 \times 6$

This result can be used to factorize trinomials? For example, to factorize the trinomial $x^2 + 7x + 12$ we need to find two numbers so that:

Product = 12 and sum = 7

The two numbers are 4 and 3

$$4 \times 3 = 12 \text{ and } 4 + 3 = 7$$

$$\therefore x^2 + 7x + 12 = (x+4)(x+3)$$

Example : Factorize i) $x^2 + 7x - 18$

$$\text{ii) } m^2 - 9m + 14$$

Solution:

$$\text{i) } x^2 + 7x - 18$$

$$\text{Product} = 9 \times (-2) = -18$$

$$\text{Sum} = 9 + (-2) = 7$$

$$\therefore x^2 + 7x - 18 = (x+9)(x-2)$$

$$\text{ii) } m^2 - 9m + 14$$

$$\text{Product} = (-7)(-2) = 14$$

$$\text{Sum} = (-7) + (-2) = -9$$

$$\therefore m^2 - 9m + 14 = (m-7)(m-2)$$

Example : Find the value of $(10001)^2$

Solution:

$$\begin{aligned}(10001)^2 &= (10000 + 1)^2 \\ &= (10000)^2 + 2(10000)(1) + (1)^2 \\ &= 100000000 + 20000 + 1 \\ &= 100020001\end{aligned}$$

Algebraic Fraction:

An expression which has a variable in the denominator, is called an algebraic expression. Algebraic fractions are added and subtracted using the same method as for arithmetic fractions. The denominator must be the same before these operations can be carried out.

Multiple Choice Questions (MCQs)

Q1. If $x = 235$ and $y = 117$, then $\frac{x^2 - y^2}{x - y} = ?$

(A) 118

(B) 100

(C) 115

(D) 352

Q2. If $x^2 - y^2 = 16$ and $x^2 + y^2 = 34$, which of the following could be the value of xy ?

- | | | |
|---|--------------------|---|
| <p>I 15</p> <p>(A) only I</p> <p>(C) I and II only</p> | <p>II -15</p> | <p>III 45</p> <p>(B) only II</p> <p>(D) III only</p> |
|---|--------------------|---|
- Q3. The average of the polynomials, $2x^2 + 5x - 6$, $5x^2 - 5x - 6$ and $30 - 7x^2$ is:
- | | |
|--------|----------|
| (A) 14 | (B) 18 |
| (C) 6 | (D) $5x$ |
- Q4. What is the value of $x^2 + 14x + 24$, when $x = 854$?
- | | |
|-------------|-------------|
| (A) 1000 | (B) 100,000 |
| (C) 741,296 | (D) 742,398 |

Explanatory Answers

Q1. (D) $\frac{x^2 - y^2}{x - y} = \frac{(x - y)(x + y)}{x - y} = x + y = 235 + 117 = 352$

Q2. (C) Adding both equations i.e.,

$$x^2 + y^2 = 34$$

$$x^2 - y^2 = 16$$

$$2x^2 = 50 \Rightarrow x^2 = 25$$

$$\Rightarrow x = \pm 25$$

$$\text{Now, } x^2 + y^2 = 34 \Rightarrow 25 + y^2 = 34 \Rightarrow y^2 = 9$$

$$\Rightarrow y = \pm 3$$

$$\text{Hence, } xy = (-5)(-3) = 15 = (5)(3)$$

$$\text{and } xy = (-5)(3) = -15 = (5)(-3)$$

So correct answer is C.

Q3. (C) First of all we find the sum of the three polynomials, then divide the answer by 3.
Sum of the three polynomials

$$\begin{array}{r} 2x^2 + 5x - 6 \\ 5x^2 - 5x - 6 \\ -7x^2 \quad + 30 \\ \hline 18 \end{array}$$

$$\text{Now, Average} = \frac{\text{Sum of the three polynomials}}{3}$$

$$= \frac{18}{3} = 6$$

Q4. (C) To avoid time consuming calculation, factorize the given polynomial

$$x^2 + 14x + 24 = x^2 + 12x + 2x + 24 = x(x + 12) + 2(x + 12)$$

$$= (x + 2)(x + 12)$$

Substituting the value of x in above

$$= (854 + 2)(854 + 12) = (856)(866)$$

$$= 741,296$$
